

Roman Penunuri

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SUMMARY

Fourth-year Electrical Engineering student with strong analytical foundations in numerical methods, signals and systems, and applied electromagnetic theory, combined with industry experience applying systematic diagnostic methodologies to complex electrical systems. Comfortable moving between simulation environments and physical measurement to validate hypotheses and design solutions. Experienced contributing to interdisciplinary technical analysis projects with real-world infrastructure applications. Self-motivated and driven to see technical problems through from start to finish, proactive about seeking input, and comfortable integrating ideas across disciplines.

EDUCATION

University of Houston, Cullen College of Engineering

Houston, TX

B.S. Electrical Engineering, Power Systems Concentration | GPA: 3.82

May 2028

Dean's List: Fall 2024, Spring 2026

Relevant Coursework: Numerical Methods for ECE, Applied Electromagnetic Waves (Honors), Signals & Systems Analysis, Engineering Statistics, Power System Analysis (in progress), Circuits and Systems, Microprocessor Systems

SKILLS

Certifications: OSHA-30 General Industry | SEL CBT 101 Protective Relay Fundamentals (4 PDH) | Pursuing ETAP Operator Certification

Proficient: MATLAB, Falstad, LTSpice, C, GE Vernova EnerVista, MS Suite

Currently Learning: Python (online coursework), ETAP, Power Electronics (MIT OCW and Coursera)

AI Tools: Proficient in AI-assisted development workflows and rapid prototyping; applied to full-stack software development from concept to working product

Languages: Fluent in English, Conversational in Spanish

PROJECTS

Water Level Indicator: BJT Sensing Circuit

University of Houston

- Designed a BJT transistor-based sensing circuit; diagnosed LED underperformance by iterating between Falstad simulation and bench oscilloscope measurements to isolate a transistor saturation failure caused by high distilled water resistivity, demonstrating hypothesis-driven debugging across simulation and physical hardware

NAE Grand Challenge: Solar Energy Storage Dashboard

University of Houston

- Led integration and built the solar economics module for a MATLAB App Designer interactive tool analyzing storage technology tradeoffs and solar infrastructure economics across global regions; synthesized findings for an interdisciplinary student engineering team

WORK EXPERIENCE

Air Liquide

Pasadena, TX

Plant Operations Electrical Engineering Intern

June 2025 - August 2025

- Diagnosed recurring motor protection failures by systematically analyzing GE Vernova Multilin 469 relay configurations, identifying parameter mismatches, and validating corrected settings in the 869 platform via EnerVista, eliminating the documented fault source and reducing projected maintenance costs by ~\$10,000
- Developed instrumentation documentation for industrial bus heater ammeter monitoring: CT wiring schematics and scope-of-work specifying measurement methodology, testing procedures, and acceptance criteria for field installation
- Revised condition-based maintenance protocols for industrial electrical assets under NFPA 70B and IEEE standards; analyzed existing time-based maintenance schedules, identified optimization opportunities, and produced documentation reviewed across multiple facilities
- Authored turnaround scope-of-work packages for electrical, battery, and motor systems; used Maximo CMMS to audit maintenance histories and reconcile workbooks, coordinating with Engineering, Safety, and Maintenance to mitigate operational losses exceeding \$1M

Take 5 Oil Change

Crosby, TX

Certified Lube Technician

March 2025 - May 2025

- Executed 40+ vehicle service cycles daily to manufacturer specifications in a fast-paced, safety-critical environment, maintaining procedural discipline and component-level precision across high-volume operations

Planet Fitness

Crosby, TX

Member Services Representative

January 2024 - March 2025

- Managed operations for a 1,000+ member facility while converting an average of 6 prospects per week through structured presentations and relationship-driven communication

ORGANIZATIONS

Tau Beta Pi Honor Society

Houston, TX

Member

April 2025 - Present

Society of Hispanic Professional Engineers (SHPE)

Houston, TX

Member

August 2023 - Present

IEEE University of Houston Chapter

Houston, TX

Member

August 2025 - Present